

X4Pro. News-2026.

by V.Zerkin, Vienna, 2026-09-04

1. Code reac1.py: universal retrieval and plotting of EXFOR data having one-string Reaction-codes, for example:

- CSP Partial cross section data
- CST Temperature dependent cross section
- ETA Average neutron yield per nonelastic event
- NU Fission-neutron yield, nu-bar
- TKE Total kinetic energy
- SIG Cross section data given with limit as DATA and DATA-MIN
- SIG Production cross section, where product coded as SF4 or in DATA (ELEM/MASS)
- PFNS Prompt-fission neutron energy spectra

-- Currently implemented search: by full-Reaction-code, DatasetID, Author1, product

2. Plotting partial cross sections from local EXFOR and remote ENDF databases

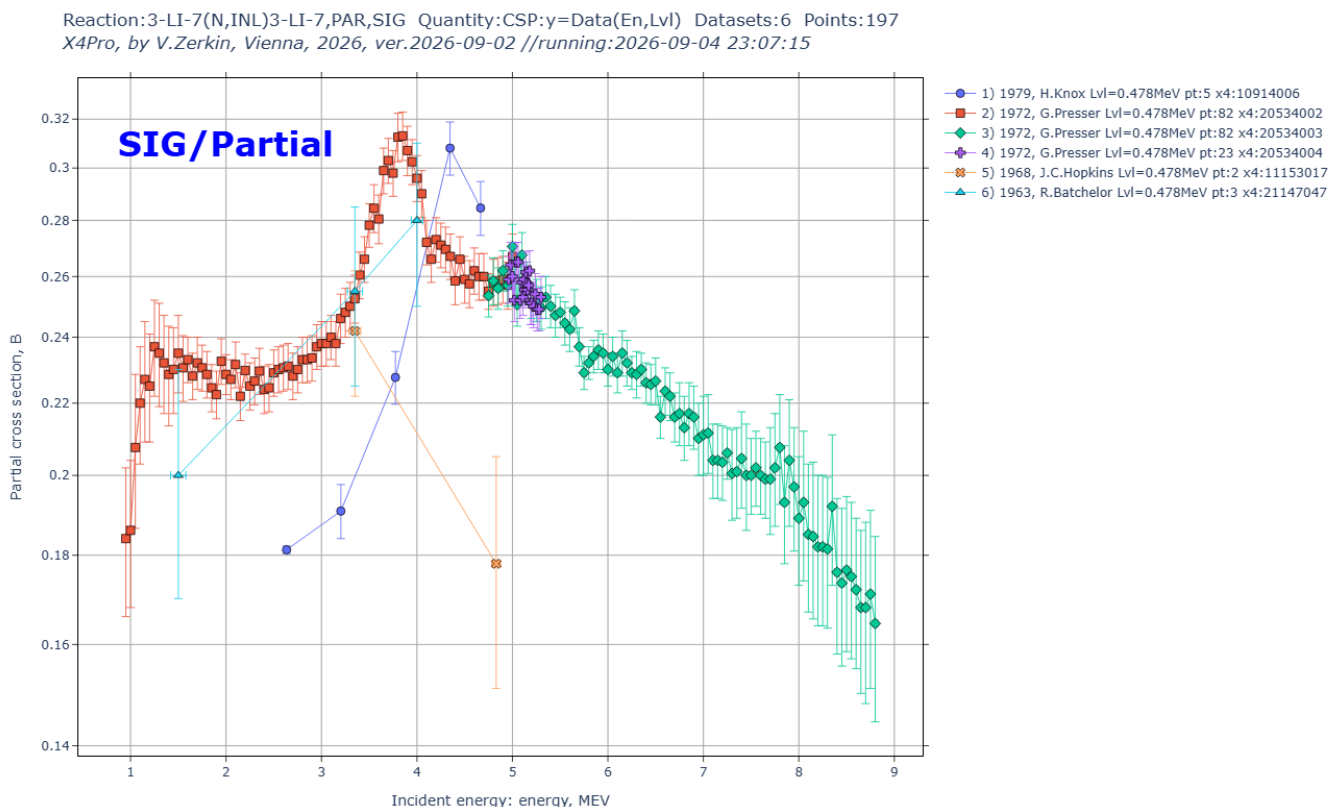
- CSP vs ENDF MT:51/MF:3,33 - Production of a neutron, with residual in the 1st excited state
- NU vs ENDF MT452,455,456/MF:1,31 - Average number of total/delayed/prompt neutrons released per fission event
- ETA vs ENDF MT:452,18,102/MF:1,3,31,33 - Average neutron yield per nonelastic event: $\eta \pm \Delta\eta$

Code reac1.py help:

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Program: reac1.py, ver.2026-08-14
Author: V.Zerkin, Vienna, 2021-2026
Purpose: Retrieve and plot any type EXFOR data from local EXFOR database
Introduction:
* computational data are presented as function y=f(x1,x2,...)
  where f is measured quantity, xi are independent variables:
      EN - incident energy,          E2 - energy of outgoing particle,
      ANG - angle of outgoing particle,  EVL - energy level, etc.
Algorithm:
* user defines EXFOR Reaction-code for data search, x-variable by number
* selects x-variable by number (default: x1)
* other variables will be used as parameter for grouping of data points
Usage:
$ python [{flag}] x4update.py [{option|reaction}]
* flag: see all Python flags: $ python --help
  -B      don't write .pyc files on import
* option:
  -help   print this help-text and exit (also --help)
  -o:<file> output file name
  -x:x?   set x variable: "x1" to "x5", e.g. -x:x2, default: -x:x1
  -x?fam:<str> filter for "families" of x? variable, e.g. -x2fam:"LVL;EXC"
  -x?min:<num> filter out values of x? variable, e.g. -x1min:6e6
  -x?max:<num> filter out values of x? variable, e.g. -x1max:30e6
  -nogrp  avoid grouping datasets by reaction-codes on the plot
  -fx:<num> set factor for x-values, e.g. -fx:1e6 sets "MeV" on X-Axis
  -fy:<num> set factor for y-values, e.g. -fy:1e-3 sets "mb/sr" on Y-Axis
plotting:
  -xlog   logarithmic scale of X-Axis
  -ylog   logarithmic scale of Y-Axis
  -lines  connect points on the plot by lines
  -sym    draw border of a symbol of a data point on the plot
  -annot:x,y,txt annotation on top of the plot;
           x and y should be given in units of the plot;
           text can include html tags: <b>, <sup>
* reaction:
  <reaction> full EXFOR Reaction-code, e.g. "8-0-16(N,EL)8-0-16,,DA"
```

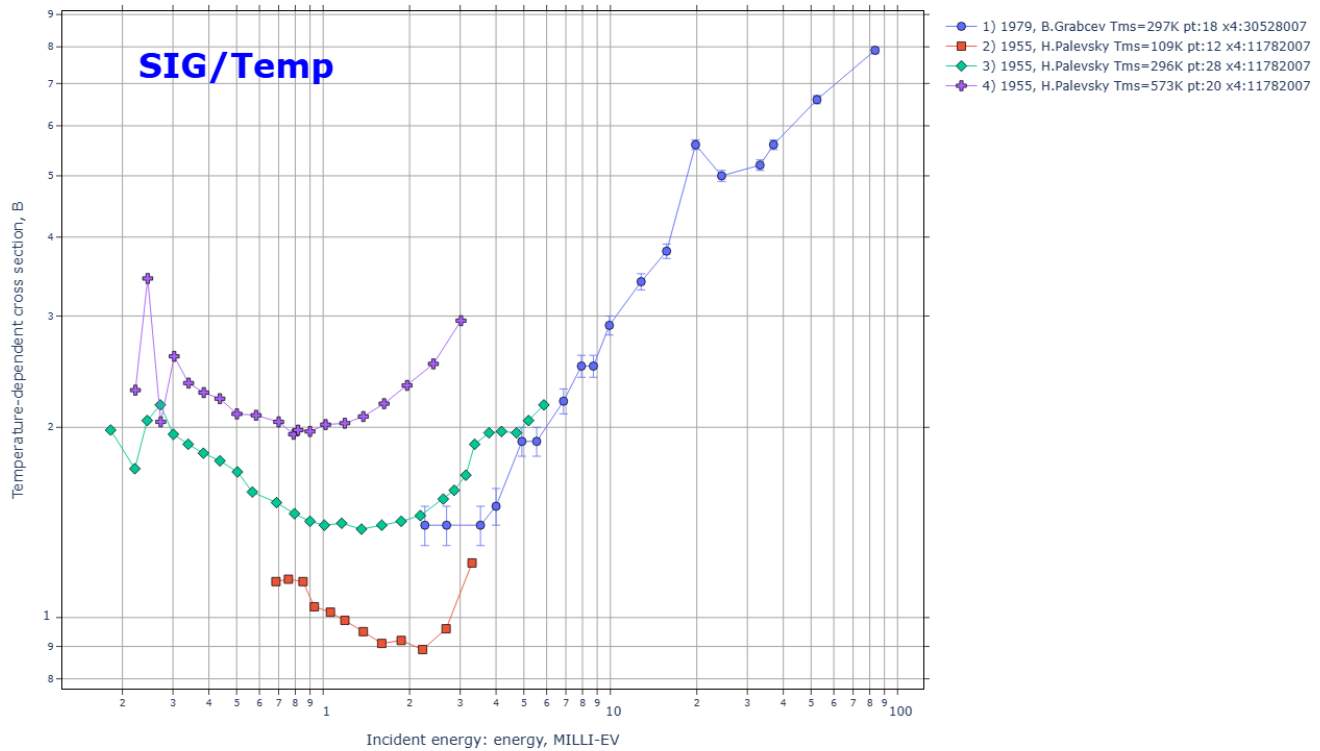
part1-8-reac/ EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA

1) part1-8-reac/out00/sig1p.html.png CSP Partial cross section data



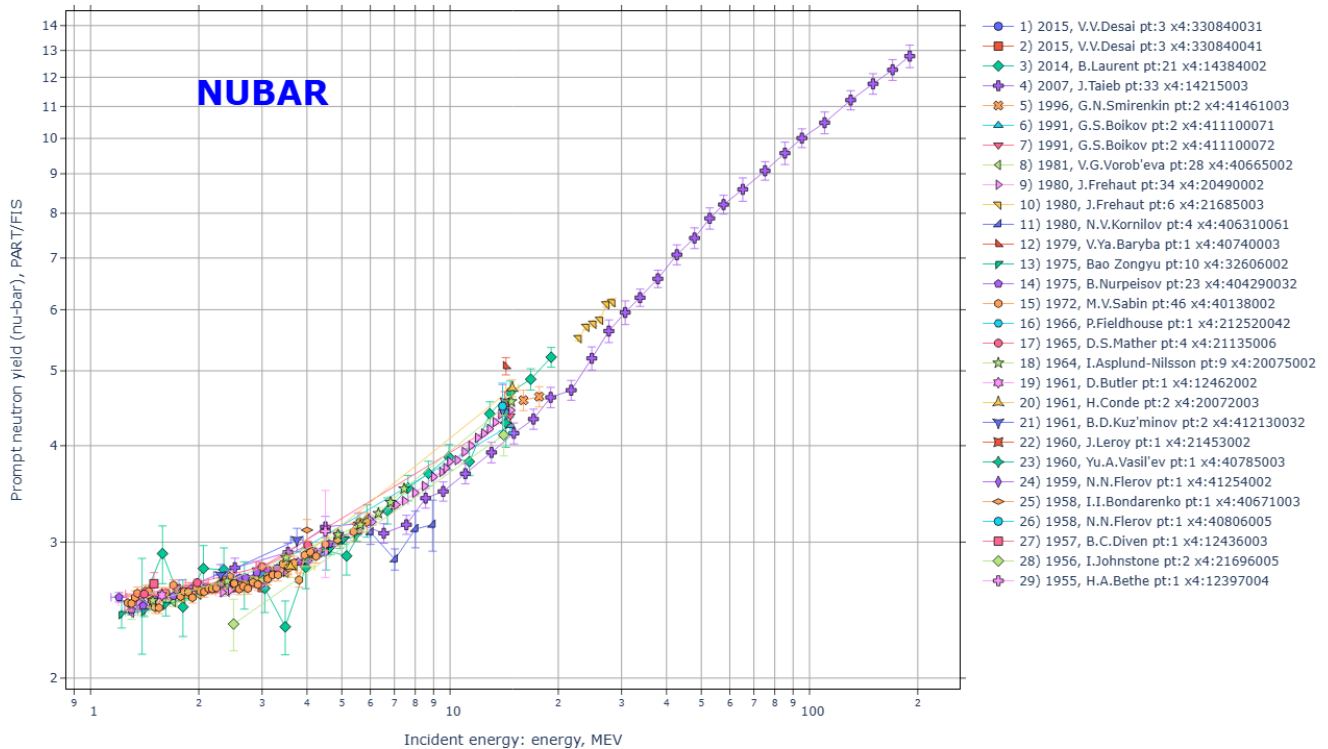
2) part1-8-reac/out00/cst1.html.png CST Temperature dependent cross section

Reaction: 82-PB-0(N,TOT),,SIG/TMP Quantity: CST:y=Data(En,Tms) Datasets: 4 Points: 78
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-04 23:07:04



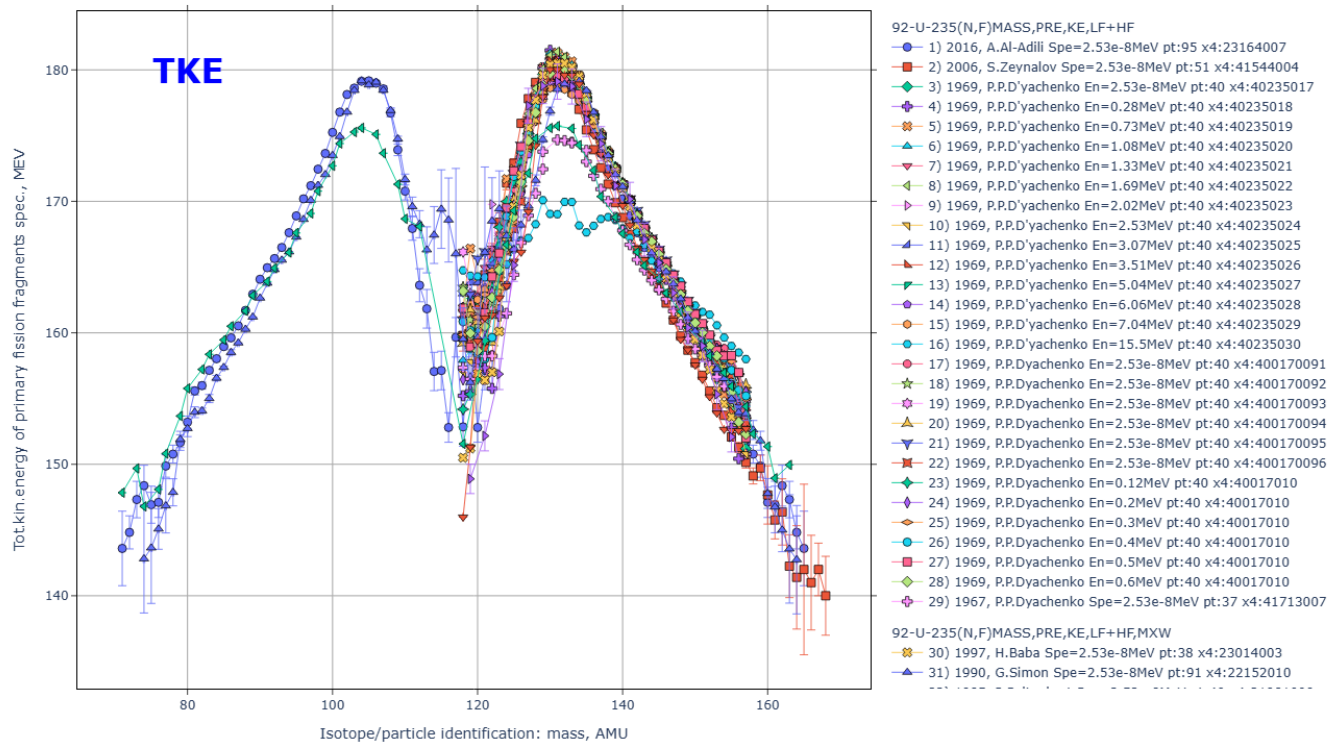
3) part1-8-reac/out00/nu1.html.png NU Fission-neutron yield, nu-bar

Reaction: 92-U-238(N,F),PR,NU Quantity: NU:y=Data(En) Datasets: 29 Points: 246
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-04 23:07:07



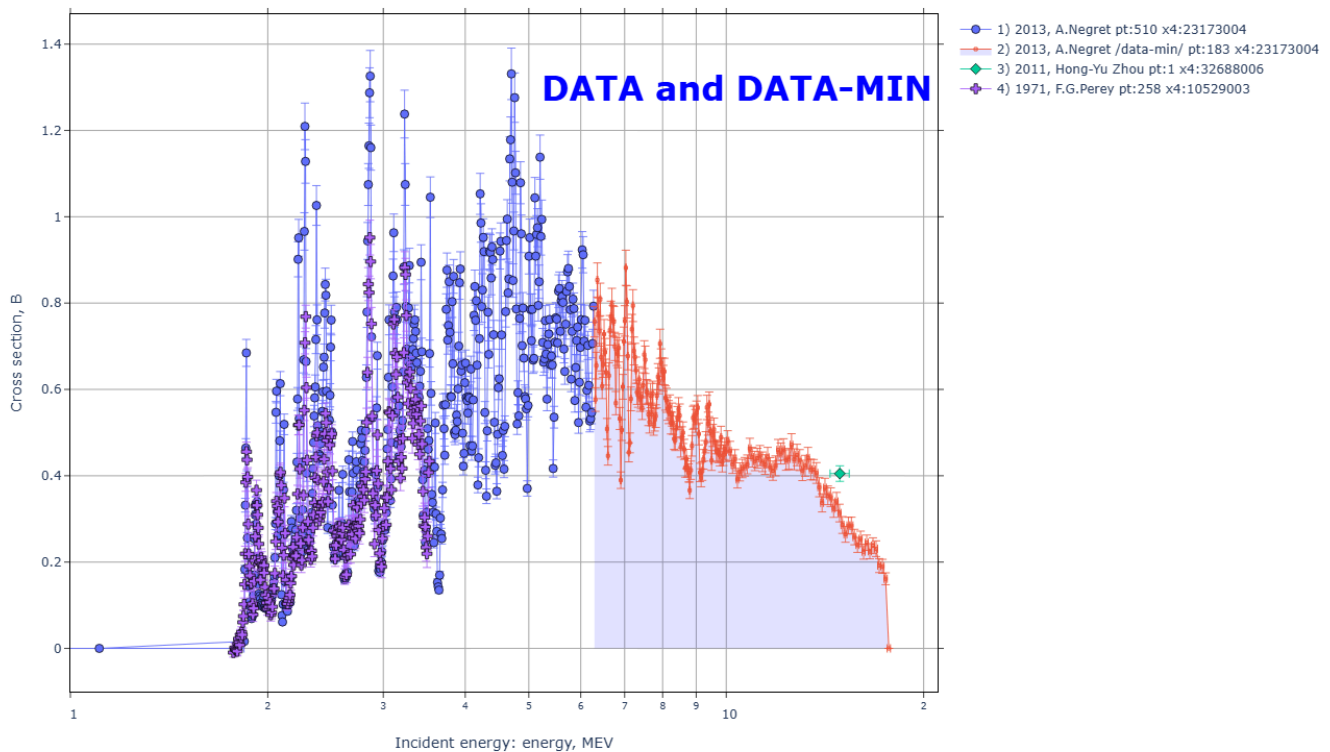
4) <part1-8-reac/out00/tke1.html.png> TKE Total kinetic energy

Reaction: 92-U-235(N,F)MASS,PRES,KE,LF+HF Quantity: E:y=Data(Spe,Zam) Datasets:34 Points:1467
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-04 23:07:09



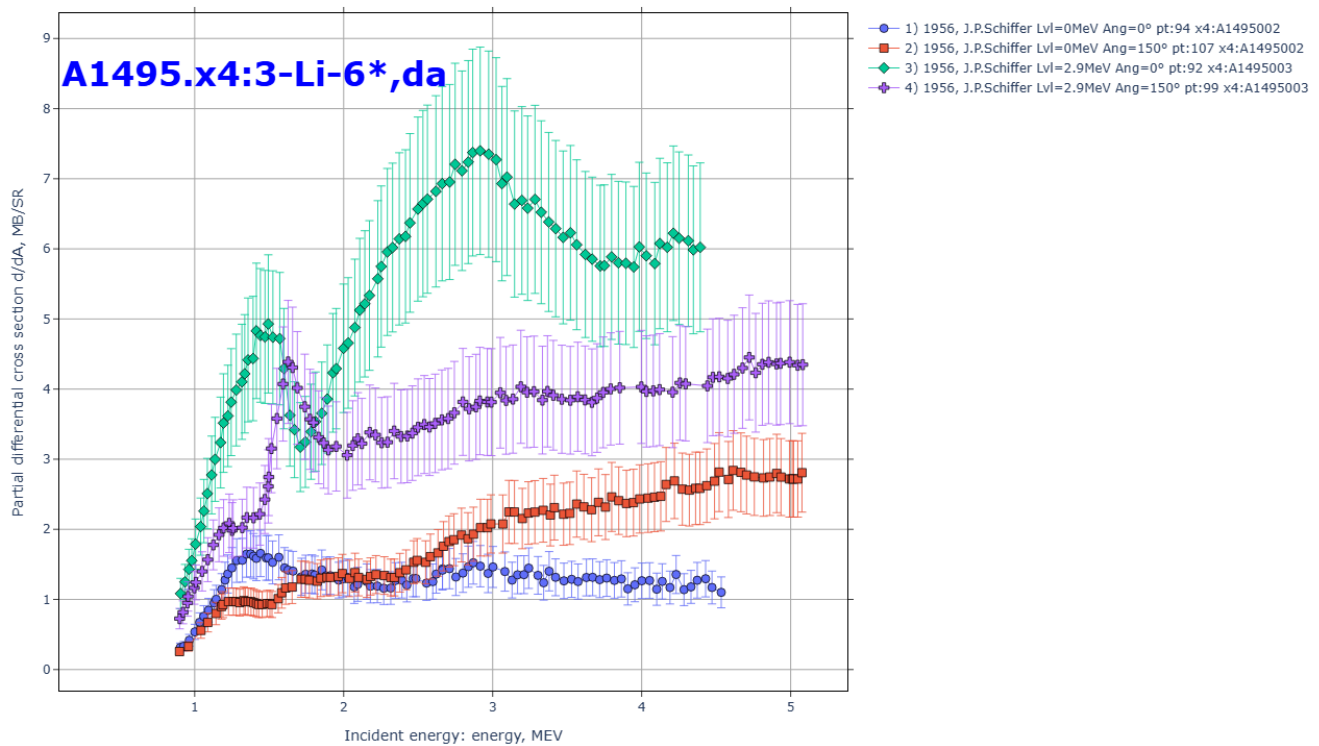
5) <part1-8-reac/out00/si28nn.html.png> SIG Cross section data given with limit as DATA and DATA-MIN

Reaction: 14-SI-28(N,INL)14-SI-28,,SIG Quantity: CS:y=Data(En) Datasets:4 Points:952
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-04 23:07:18



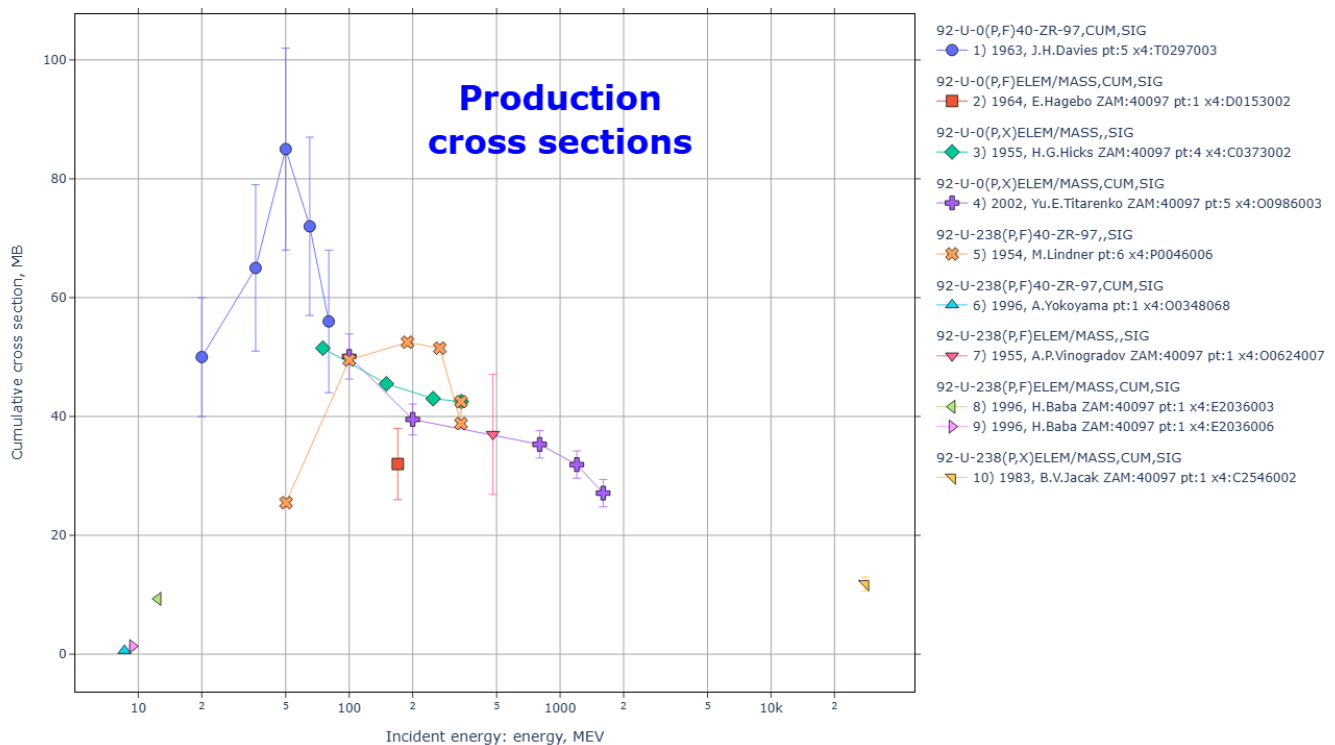
6) <part1-8-reac/out00/a1495-li6.html.png> Data from EXFOR:"A1495" & Reaction: "3-LI-6*DA"

Reaction: 3-LI-6(HE3,P)4-BE-8,PAR,DA Quantity: DAP:y=Data(En,Lvl,Ang) Datasets: 4 Points: 392
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-04 23:07:21



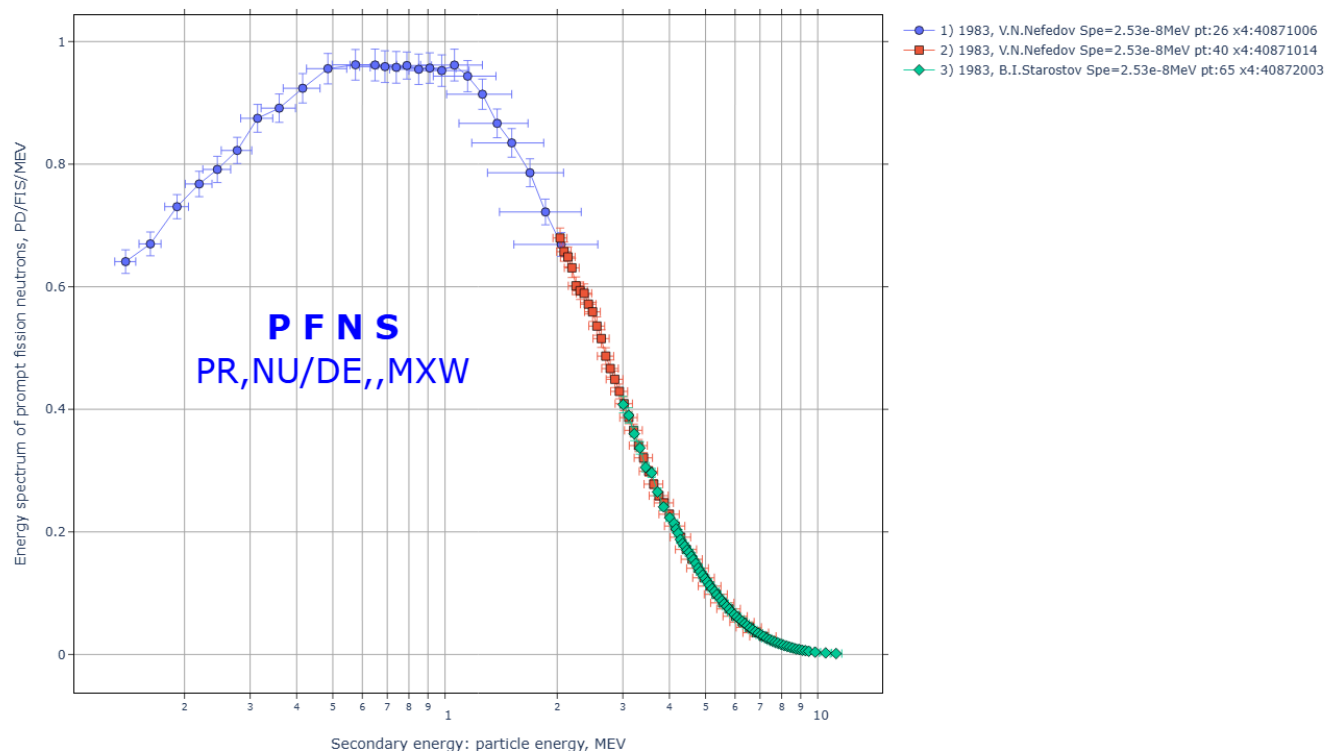
7) <part1-8-reac/out00/sig1prod.html.png> Production cross section coded as SF4 and ELEM/MASS

Reaction: 92-U-0(P,F)40-ZR-97,CUM,SIG Quantity: CS:y=Data(En) Datasets: 10 Points: 26
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-04 23:07:29



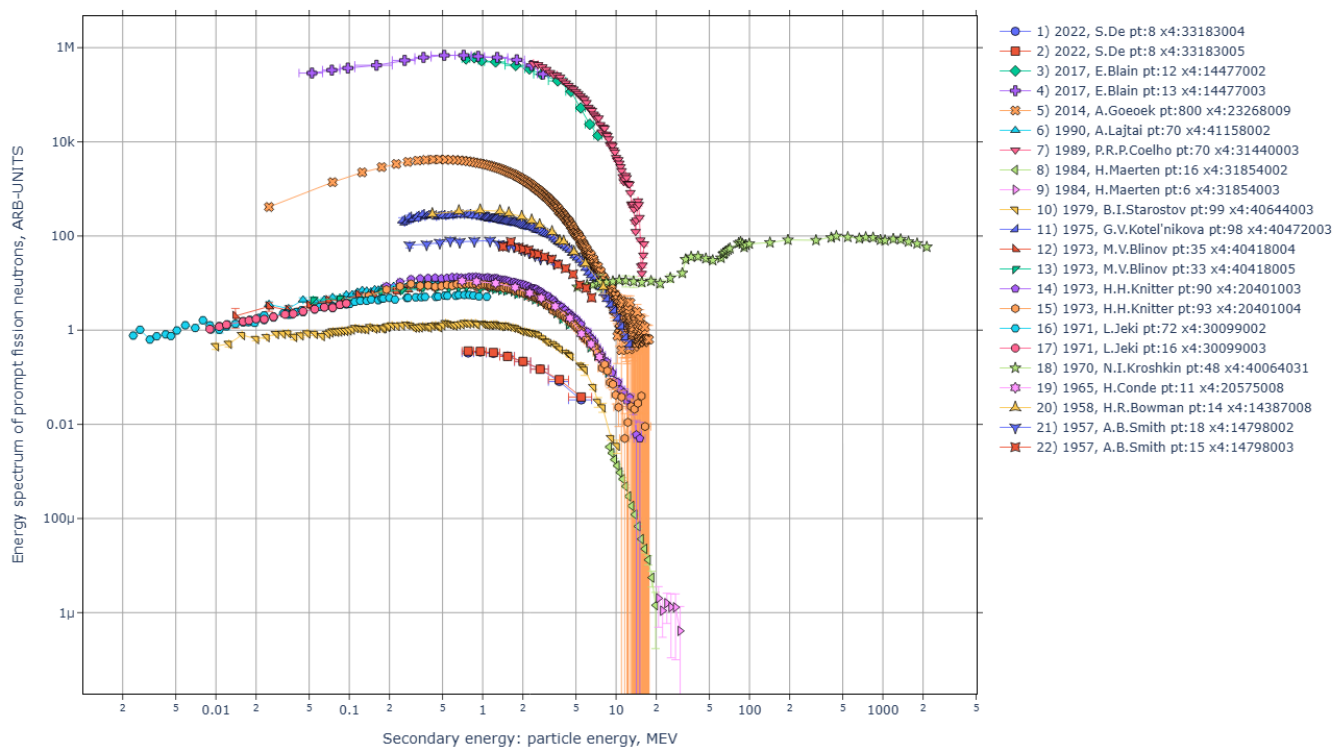
8) [part1-8-reac/out00/pfns1.html.png](#) PFNS Prompt-fission neutron energy spectra

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXW Quantity: MFQ:y=Data(Spe,E2) Datasets: 3 Points: 131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-04 23:07:32

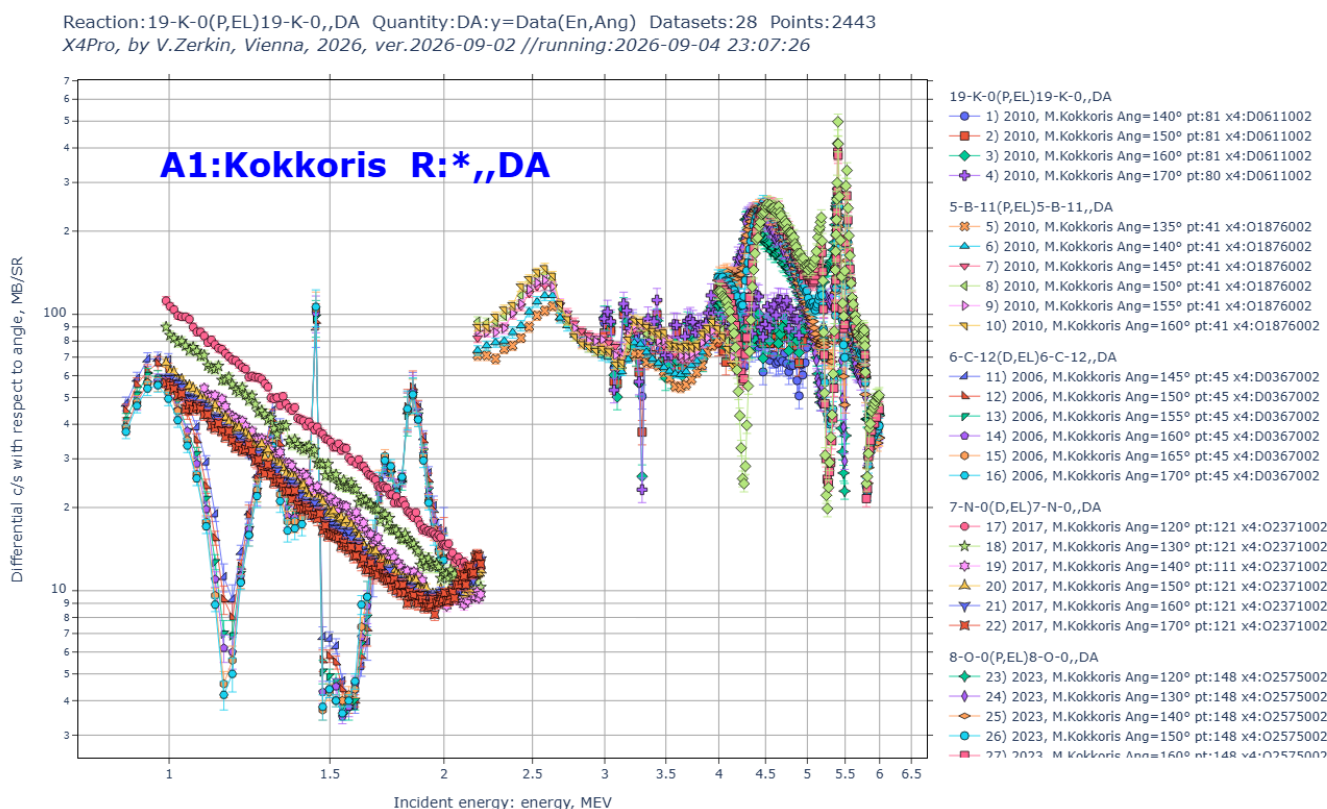


9) [part1-8-reac/out00/pfns4.html.png](#) Energy spectrum of prompt fission neutrons: relative data

Reaction: 98-CF-252(0,F),PR,NU/DE,,REL Quantity: MFQ:y=Data(E2) Datasets: 22 Points: 1645
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-04 23:07:39

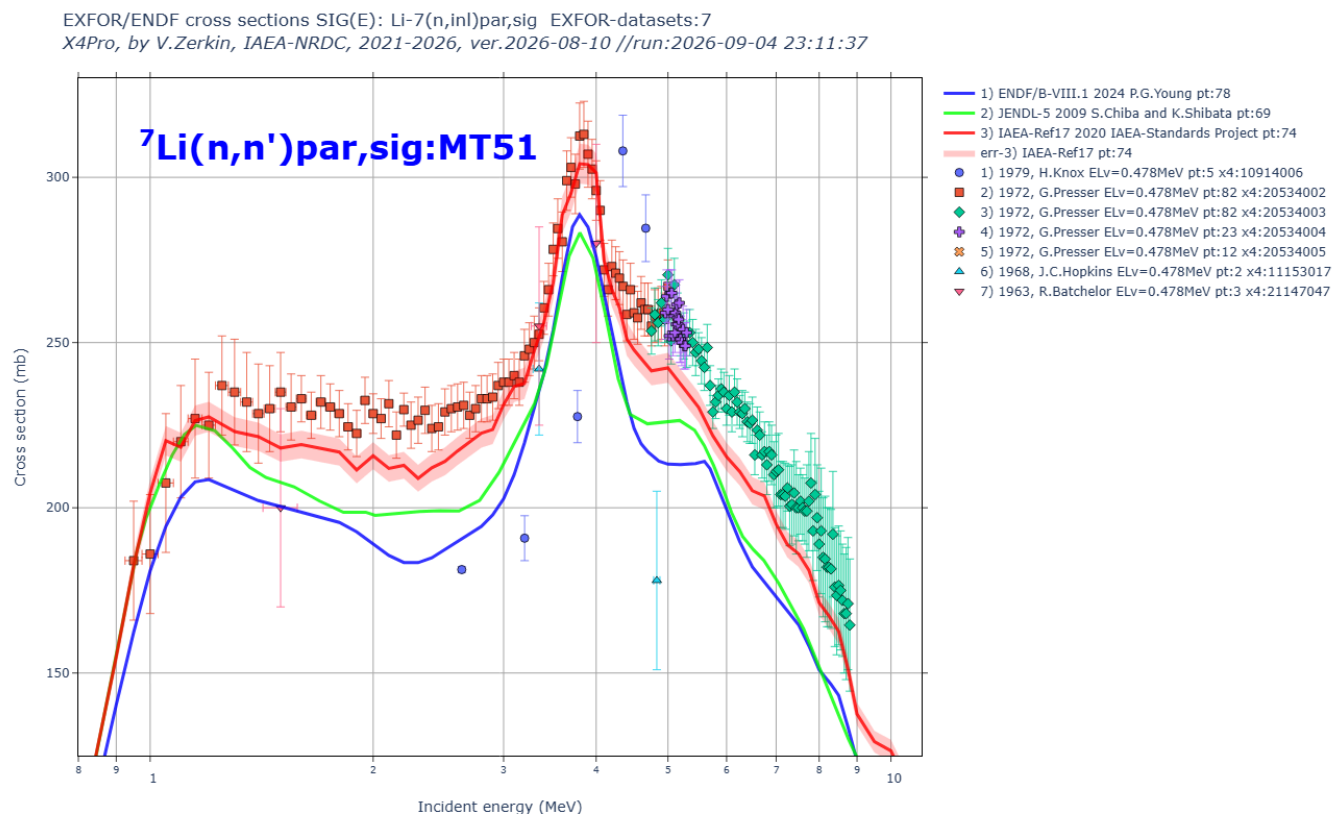


10) part1-8-reac/out00/Kokkoris.html.png Data by "Kokkoris" & Reaction: "*,DA"



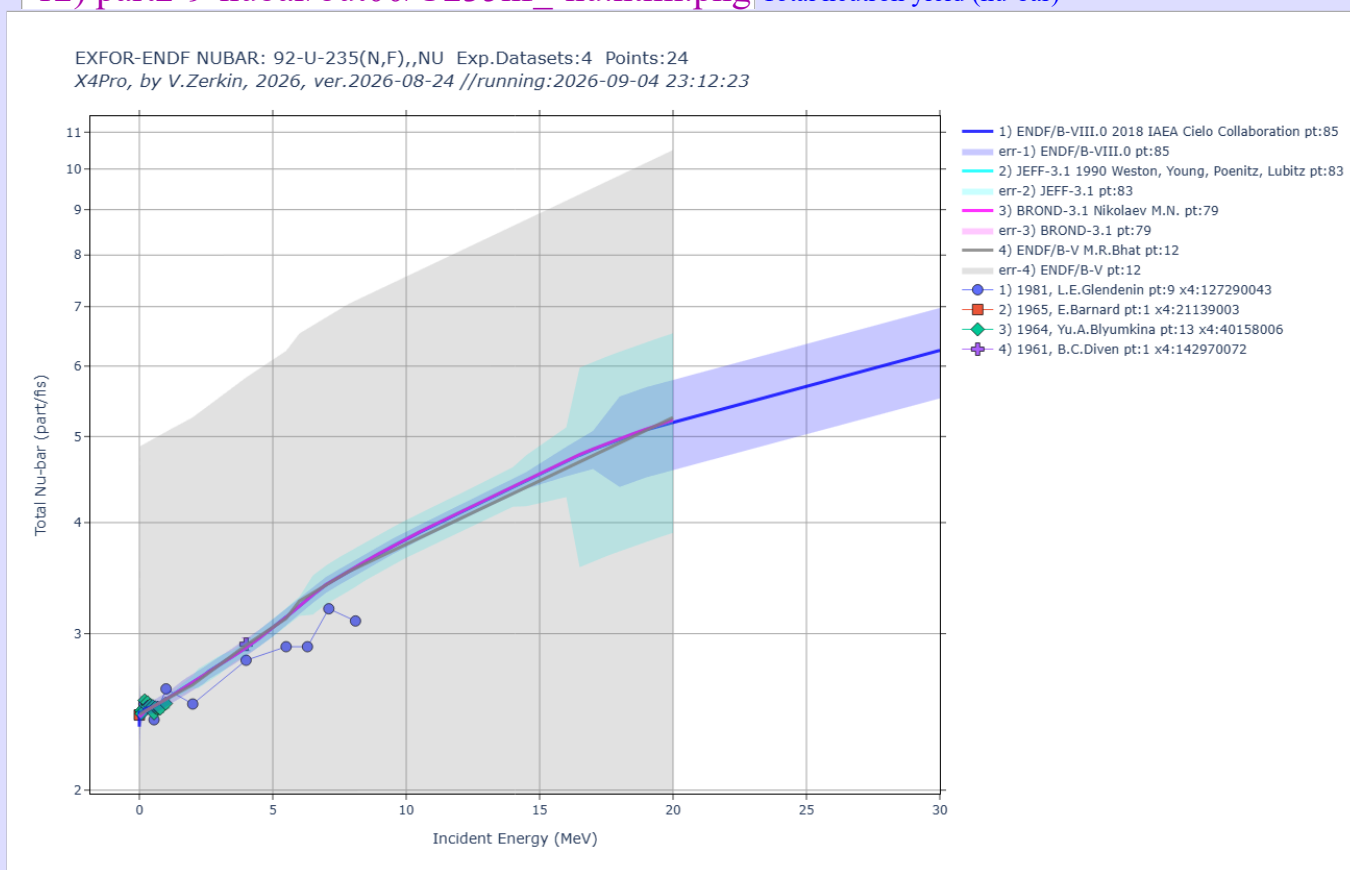
part2-8-sig1par/ EXFOR/ENDF partial cross sections CSP:MT51

11) part2-8-sig1par/out00/Li7ninl.html.png Production of a neutron, with residual in the 1st excited state

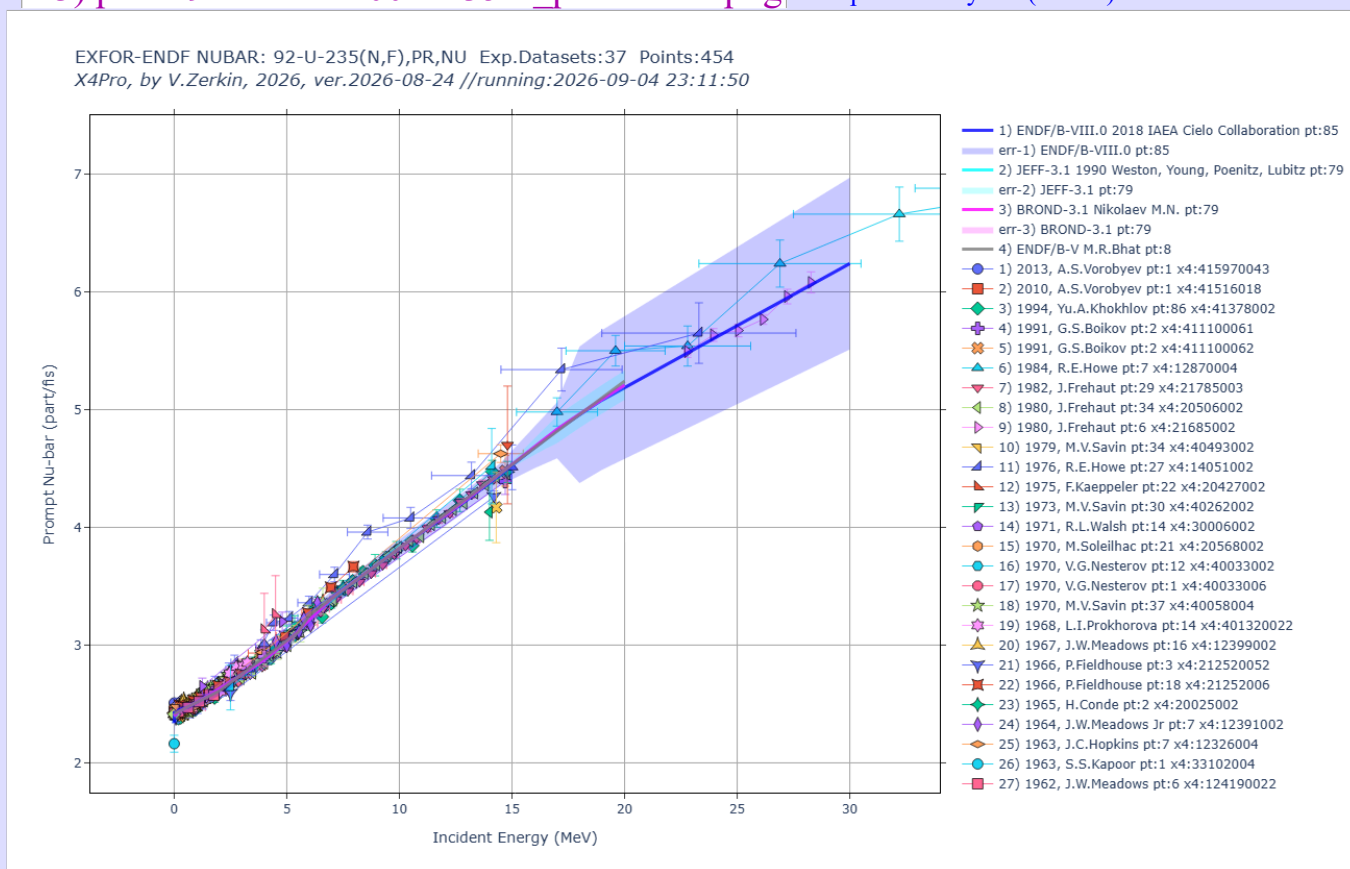


part2-9-nubar/ EXFOR/ENDF NUBAR - average number of neutrons per fission

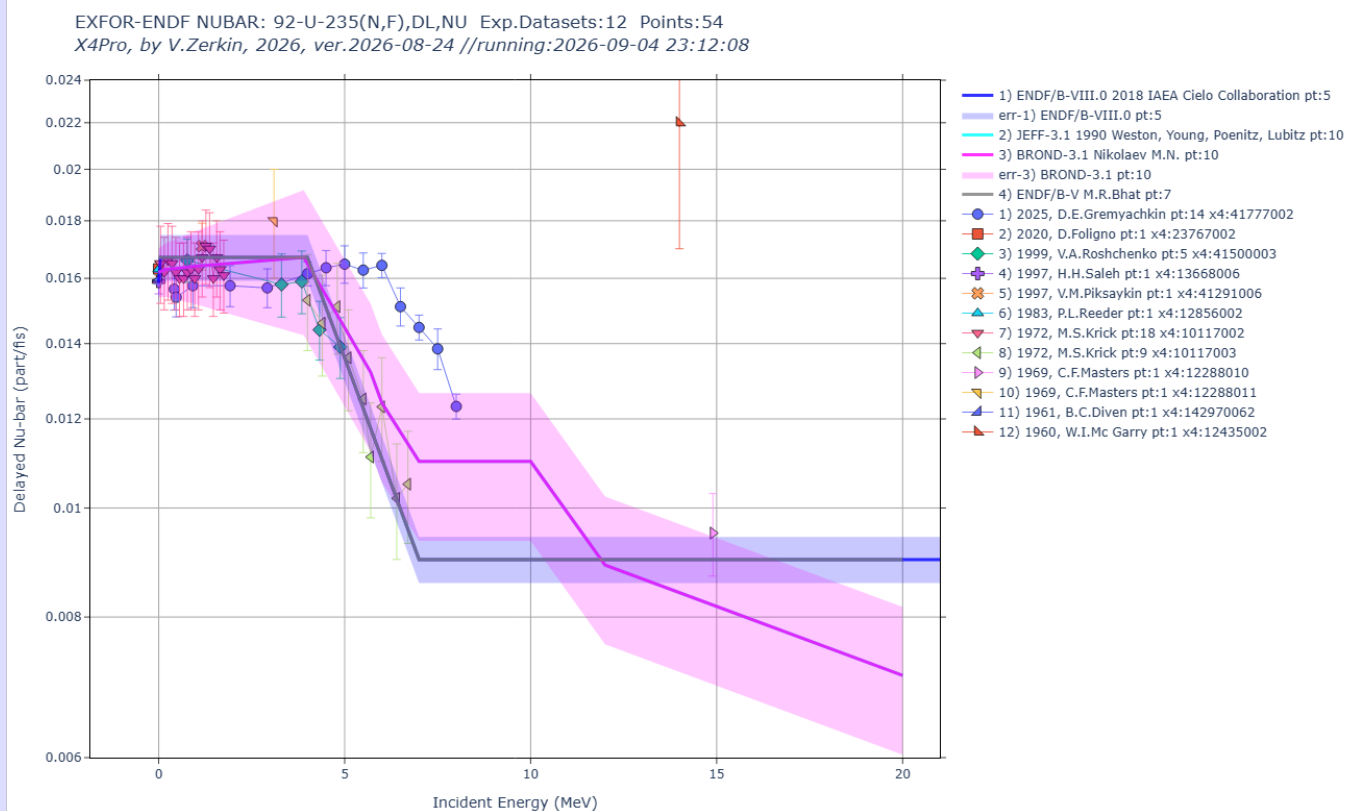
12) part2-9-nubar/out00/U235nf_-nu.html.png Total neutron yield (nu-bar)



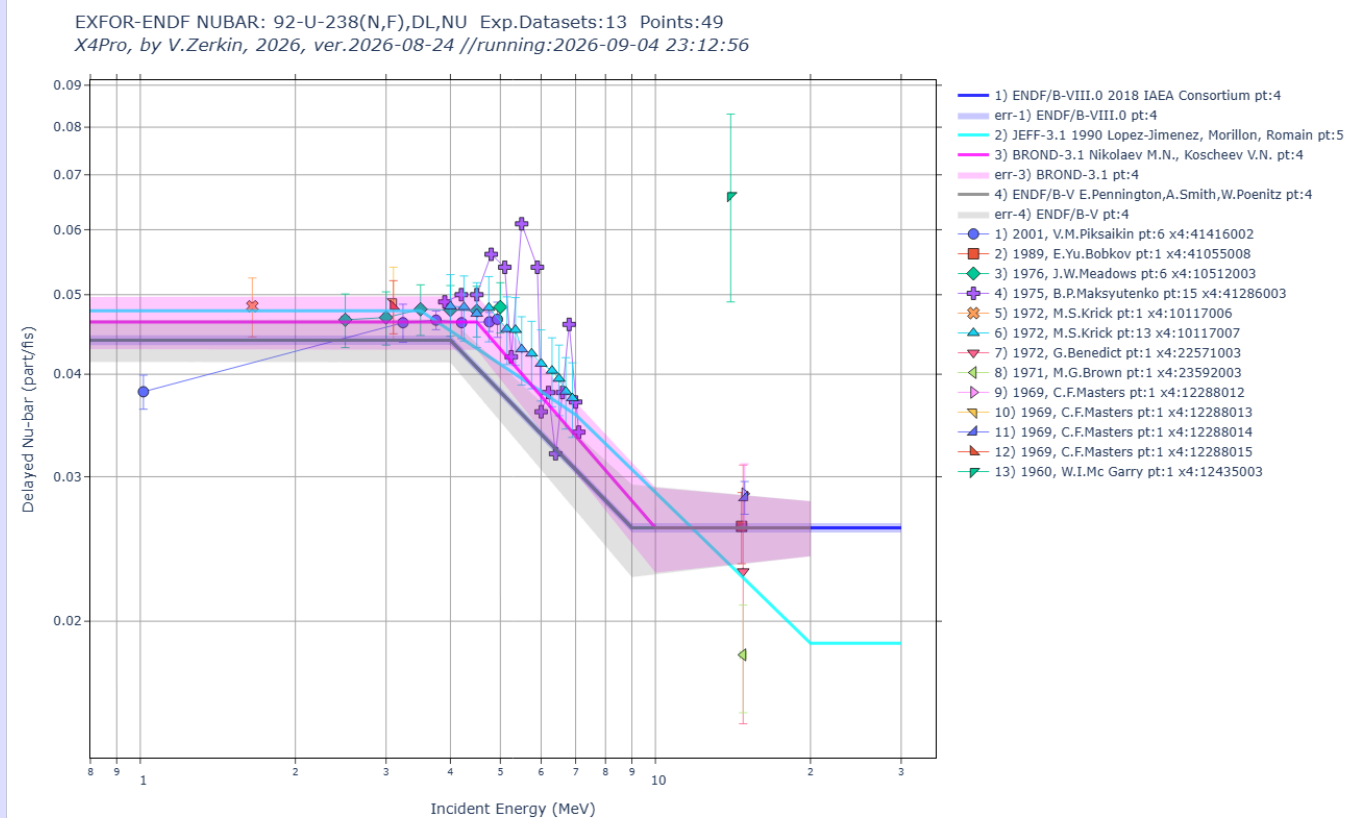
13) part2-9-nubar/out00/U235nf_pr-nu.html.png Prompt neutron yield (nu-bar)



14) [part2-9-nubar/out00/U235nf_dl-nu.html.png](#) Delayed neutron yield



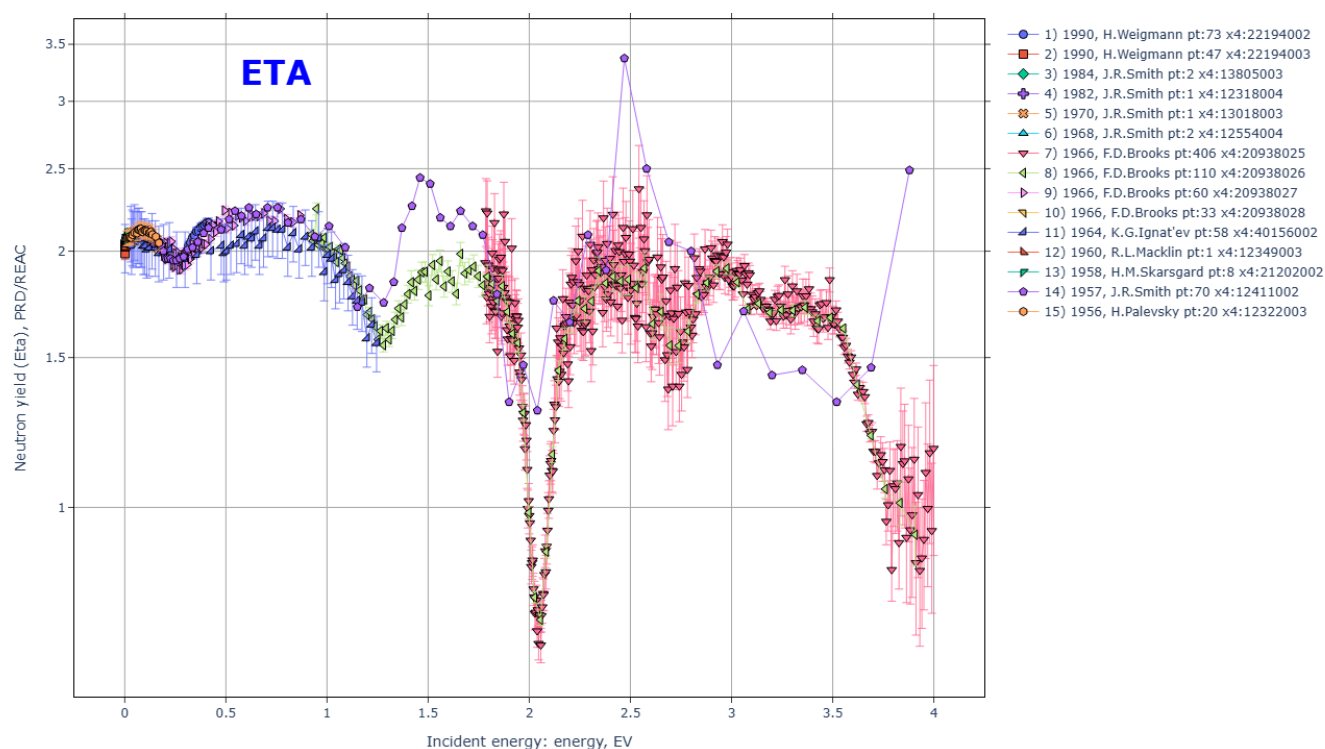
15) [part2-9-nubar/out00/U238nf_dl-nu.html.png](#)



part1-8-reac/ EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA

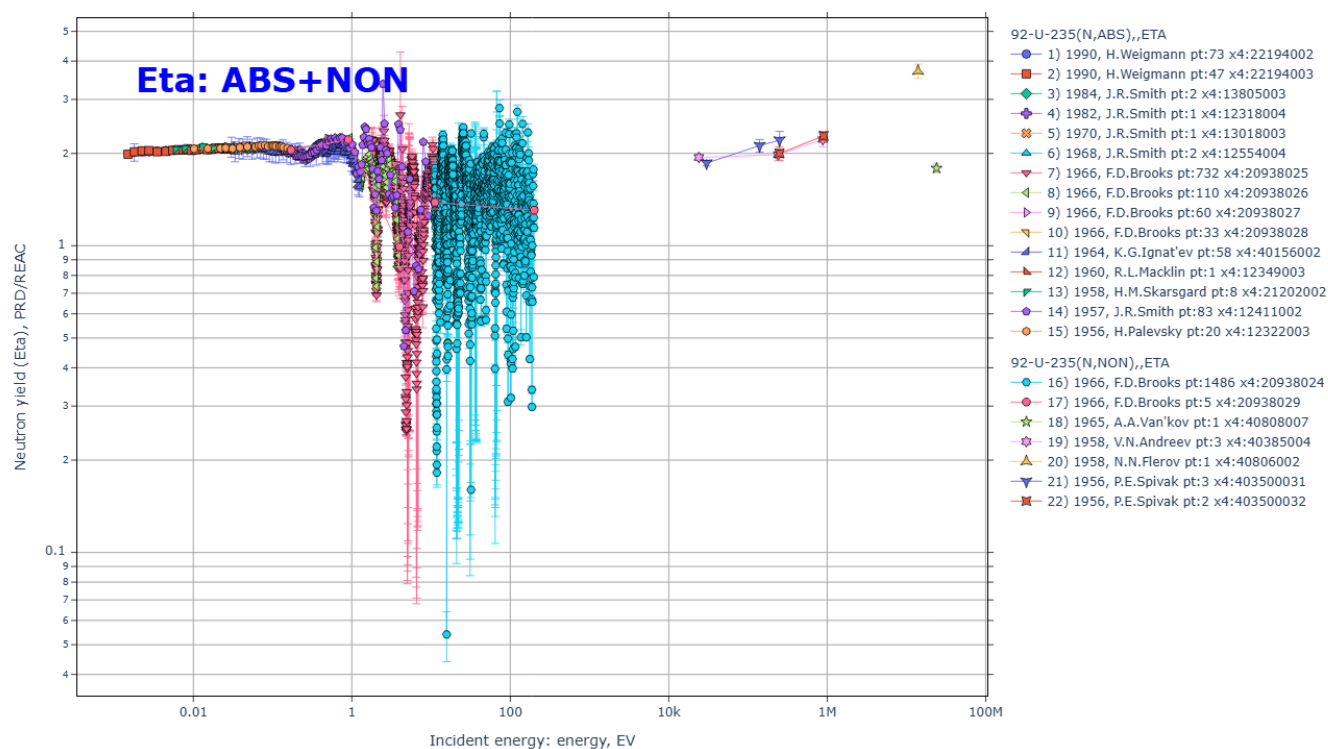
16) part1-8-reac/out00/eta1.html.png ETA Average neutron yield per nonelastic event

Reaction:92-U-235(N,ABS),,ETA Quantity:NU:y=Data(En) Datasets:15 Points:892
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-08-28 13:26:55

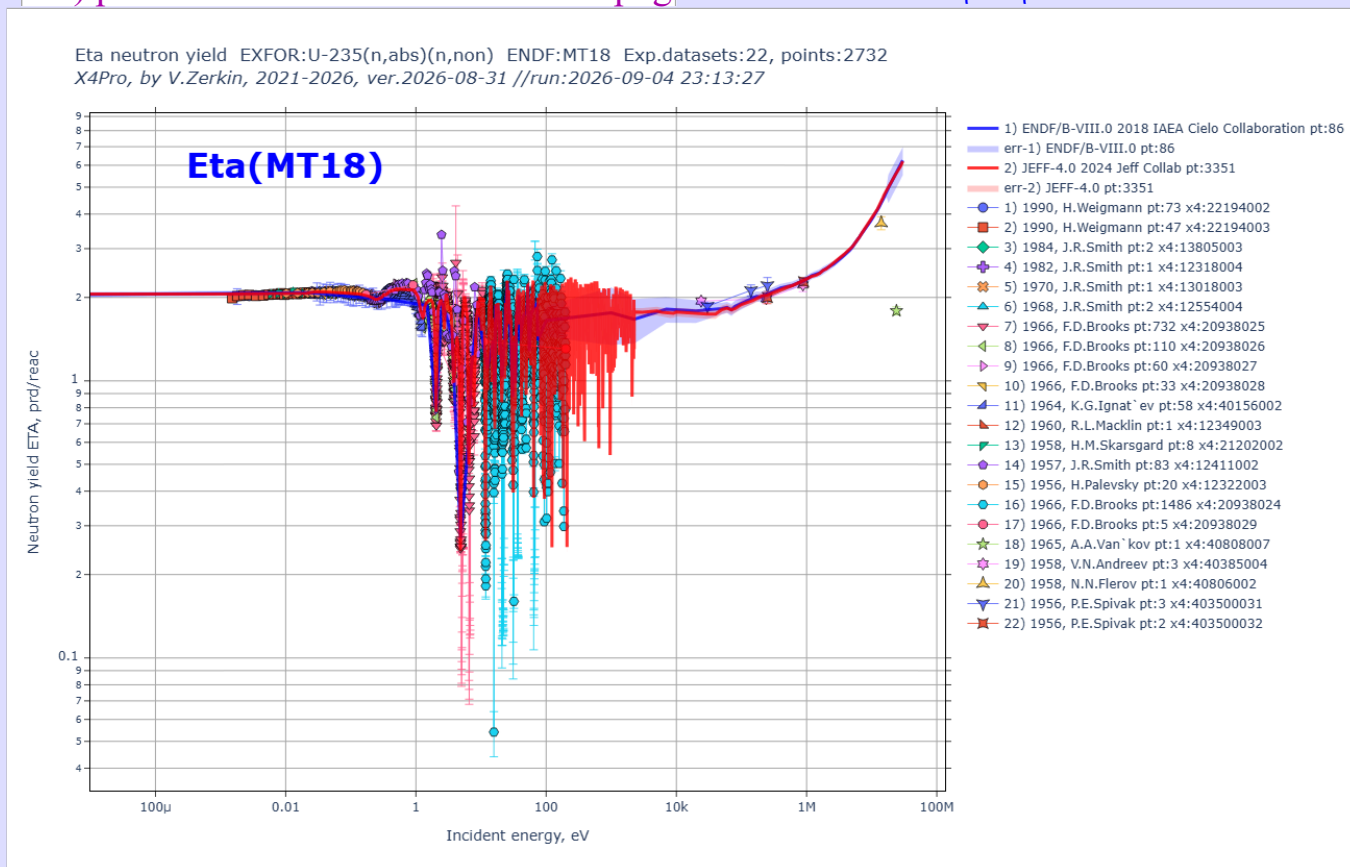


17) part1-8-reac/out00/eta2.html.png ETA: (N,ABS)+(N,NON)

Reaction:92-U-235(N,ABS),,ETA Quantity:NU:y=Data(En) Datasets:22 Points:2732
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-02 //running:2026-09-04 23:07:12



part2-A-eta/ EXFOR/ENDF Eta: neutron yield per nonelastic even

18) part2-A-eta/out00/U235-eta.html.png ETA: EXFOR vs. ENDF $\eta \pm \Delta\eta$ 

19) myplot1.png Interactive plot: EXFOR:ETA + ENDF:MT:452,18,102,MF:1,3,31,33

X4Pro - advanced plot
by V.Zerkin, 2025-2026